PENETRATION TESTING REPORT

RECONNAISSANCE & FOOTPRINTING

Field	Detail
Pentester Name (Cybersecurity Professional)	Prathamesh Kisan Satre
Program/Batch	B084B – Networkwalks
Date	10 November 2026
Modules completed	W2-PM1 (Multiple Kali Linux Tools) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (networkwalks.com) 2. My own local LAN Network
Permission secured from client?	Yes – for authorized practical scope
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery

1. Liability Disclaimer

The activities mentioned in this report were performed for educational and cybersecurity learning purposes. Testing was limited to the target and environment covered under the approved practical scope.The main objective of these exercises was to understand how publicly available information about a website can be collected using different reconnaissance tools.

No exploitation or unauthorized access was attempted during these activities. The information collected is intended only for learning and security assessment purposes.Any practical cybersecurity activity should be performed with proper authorization. The misuse of these techniques may result in legal consequences.

2. Introduction

Reconnaissance and footprinting are important initial stages of penetration testing. Before performing any detailed security assessment, it is necessary to understand the target website, its infrastructure and the information it makes publicly available.In this practical, I performed different reconnaissance activities on the Networkwalks website using Kali Linux. I used multiple command-line tools to collect information related to domain registration, DNS configuration, website technologies and HTTP responses.

Each tool helped me understand a different aspect of the target. By combining the results, I was able to develop a basic understanding of the website's publicly visible infrastructure.The following sections explain the tools used, the activities performed and the observations obtained during the practical.

3. Tools Used

Tool	**Purpose**
WHOIS	Collects publicly available domain registration information.
cURL	Checks HTTP response headers from a website.
DNSRecon	Performs DNS record enumeration.
Nslookup	Finds the IP address associated with a domain name.
Wafw00f	Identifies possible Web Application Firewall technology.
WhatWeb	Detects technologies and software used by websites.
Kali Linux	Platform used to execute reconnaissance commands.

4. Activities Performed

4.1 WHOIS – Domain Information Gathering

I started the practical by performing a WHOIS lookup on the target domain, networkwalks.com.WHOIS is a command-line utility that provides publicly available information about domain registration. It can help identify the registrar, registration details, domain status and name servers.

After executing the command, I received domain-related registration information. The output also displayed information about the domain's name servers.During the lookup, a rate-limit notification was displayed, which restricted some of the information returned by the service.

Observation: The WHOIS command helped me understand how domain registration information can be collected during the initial reconnaissance stage.

Security Relevance: Public registration information may help security professionals understand the domain's administrative and infrastructure details.

4.2 cURL – HTTP Header Analysis

After completing the WHOIS lookup, I used cURL to examine the HTTP response headers of the target website. The `-I` option was used to retrieve only the response headers instead of downloading the complete webpage. The response displayed an HTTP/2 200 status code, indicating that the server successfully processed the request. The response headers also provided technical information, including the Apache server field and other HTTP-related details.

Observation: The command provided useful information about how the web server responds to incoming HTTP requests.

Security Relevance: HTTP headers may expose technical details that can help identify the technologies running behind a website. Organizations should review unnecessary information disclosed through server responses.

4.3 DNSRecon – DNS Enumeration

I then performed DNS enumeration using DNSRecon against networkwalks.com.

DNSRecon is a useful reconnaissance tool for collecting publicly available DNS information. It helps identify different DNS records associated with a domain. During execution, the tool returned DNS-related information, including name-server and mail/service-related records.

This activity provided a better understanding of the domain's DNS configuration and the services associated with it.

Observation: DNSRecon successfully collected DNS-related information and helped build a basic profile of the target domain.

Security Relevance: DNS information can provide useful details about an organization's internet-facing infrastructure. Regular DNS record reviews help identify outdated or unnecessary entries.

4.4 Nslookup – IP Address Identification

Next, I used the Nslookup command to check the IP address associated with the target domain.

Nslookup is a network administration tool that sends DNS queries and displays the corresponding domain resolution information.

The command successfully resolved networkwalks.com and returned the following IPv4 address:

Resolved IP Address: 192.232.216.135

The output also displayed the DNS resolver used to perform the lookup.

Observation: The domain name was successfully translated into its corresponding IP address.

Security Relevance: IP address identification is an important part of reconnaissance because it helps security professionals understand the network location of publicly accessible services.

4.5 Wafw00f – Web Application Firewall Detection

In this activity, I used Wafw00f to check whether a Web Application Firewall was present in front of the target website. A Web Application Firewall (WAF) monitors HTTP requests and helps protect web applications against different types of malicious traffic.

After executing the command, Wafw00f identified the following WAF technology:

Detected WAF: ModSecurity (SpiderLabs)

This result indicated that the website was associated with ModSecurity protection.

Observation: The tool helped identify the possible security technology protecting the web application.

Security Relevance: Identifying a WAF helps an analyst understand the defensive mechanisms used by a website. However, the presence of a WAF does not guarantee complete protection against every web-based attack.

4.6 WhatWeb – Website Technology Identification

Finally, I used WhatWeb to identify the technologies used by the target website.

WhatWeb is a website fingerprinting tool that examines web responses to identify technologies such as web servers, content management systems and other application components. The output displayed Apache-related server information along with WordPress-related technology indicators. The tool also returned additional information about the website based on the responses received during the scan.

Observation: WhatWeb helped identify technologies that were publicly detectable through the website's responses.

Security Relevance: Technology fingerprinting helps security professionals understand the software components used by a website. These components can then be reviewed for available updates and known security issues.

5. Risk Analysis / Impact

Based on the information collected during the practical, I identified the following observations and their possible security implications.

No.	Finding	Evidence / Observation	Possible Impact	Risk
1	Domain registration information available	WHOIS returned registration-related details	Helps build a profile of the target domain	Low
2	Web server information exposed	cURL displayed Apache-related server information	May assist technology identification	Low
3	DNS records publicly available	DNSRecon returned DNS-related records	Helps understand the domain infrastructure	Low
4	Public IP address identified	Nslookup returned 192.232.216.135	Reveals the website's public network address	Informational
5	WAF technology identified	Wafw00f detected ModSecurity (SpiderLabs)	Provides information about defensive technology	Informational
6	Website technologies detected	WhatWeb displayed Apache and WordPress-related indicators	May help identify software components requiring review	Low

Note: These are preliminary observations from reconnaissance activities. They do not confirm the existence of security vulnerabilities. Additional authorized testing would be required to establish whether any actual security weakness exists.

6. Recommendations

Based on the observations made during the practical, I would recommend the following security measures:

1. **Limit unnecessary information disclosure:** Review publicly visible website information and avoid exposing technical details that are not required.
2. **Keep website software updated:** Regularly update the web server, WordPress installation, plugins and other related components.
3. **Review HTTP response headers:** Check server response headers and configure appropriate security headers.
4. **Maintain proper DNS configuration:** Periodically review DNS records and remove outdated or unnecessary entries.
5. **Configure WAF protection properly:** Ensure that ModSecurity is maintained, configured correctly and monitored for suspicious activity.
6. **Monitor website technologies:** Regularly review the technologies used by the website and check for security advisories.
7. **Perform regular security assessments:** Conduct authorized security checks to identify potential weaknesses before they can be misused.

7. Conclusion

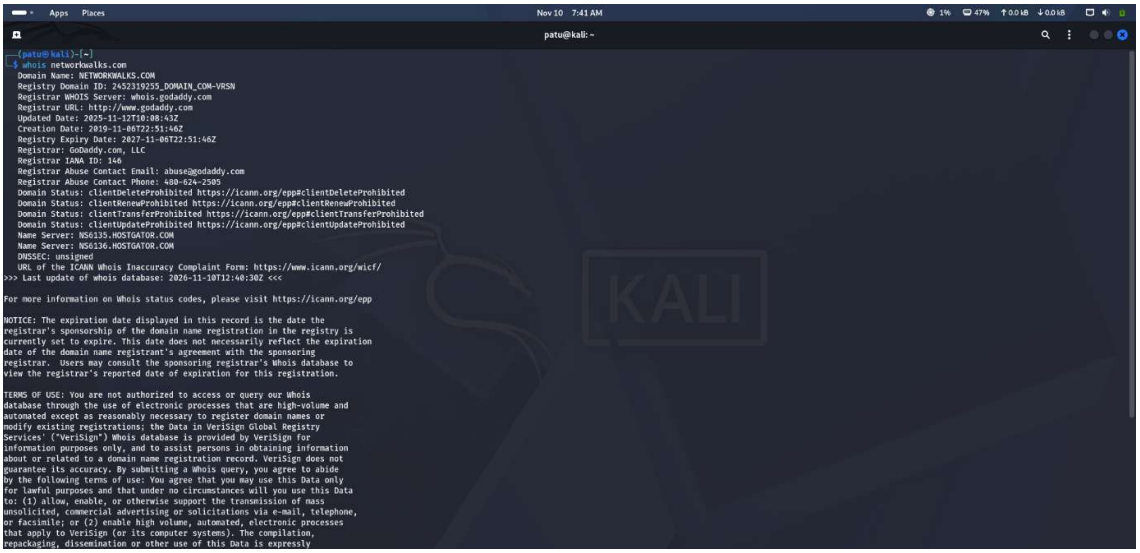
During this practical exercise, I learned how reconnaissance and footprinting help in understanding the publicly available information associated with a website. I used six different tools in Kali Linux, namely WHOIS, cURL, DNSRecon , Nslookup, Wafw00f and WhatWeb. Each tool provided different information about the target domain. Through WHOIS, I learned how to collect domain registration information. cURL helped me examine HTTP response headers, while DNSRecon and Nslookup provided information about DNS records and domain resolution.

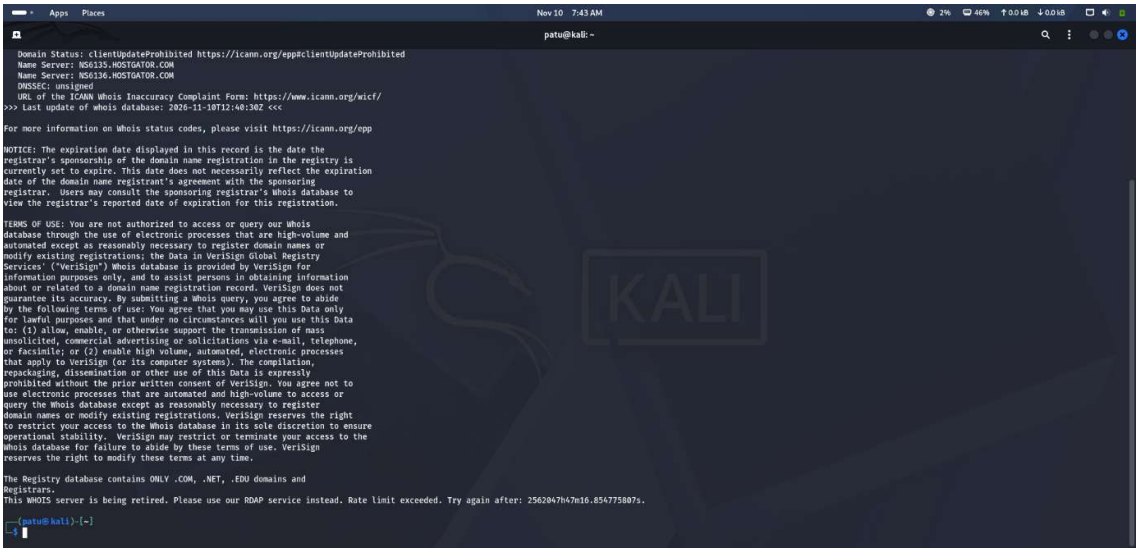
I also understood how Wafw00f can identify Web Application Firewall technology and how WhatWeb can detect website technologies. Overall, this practical helped me understand the importance of information gathering before conducting further penetration testing activities. I also learned that reconnaissance findings must be properly documented and carefully analyzed instead of directly treating every observation as a vulnerability.

8. Evidences Collected

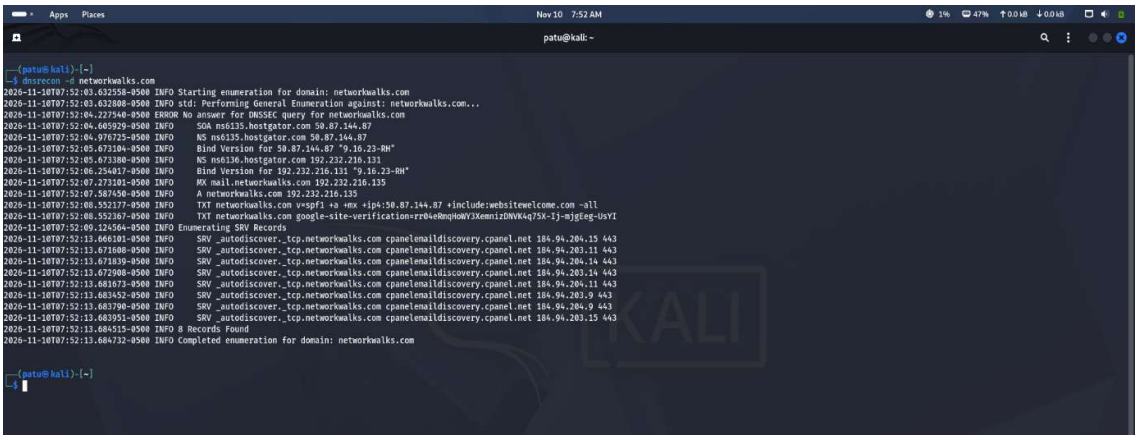
The following practical screenshots are attached as evidence of the activities performed:

1. WHOIS command execution and output.





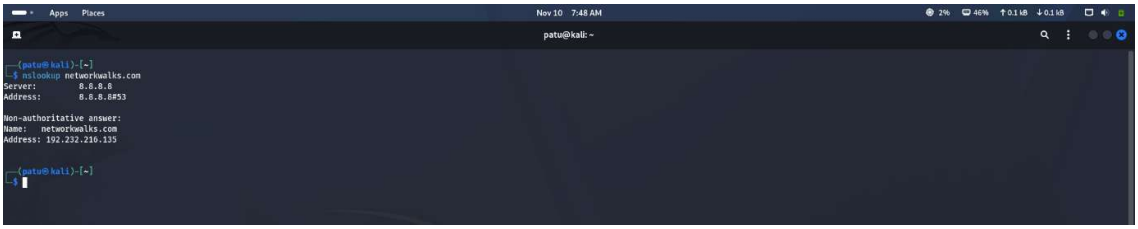
2. DNSRecon enumeration results.



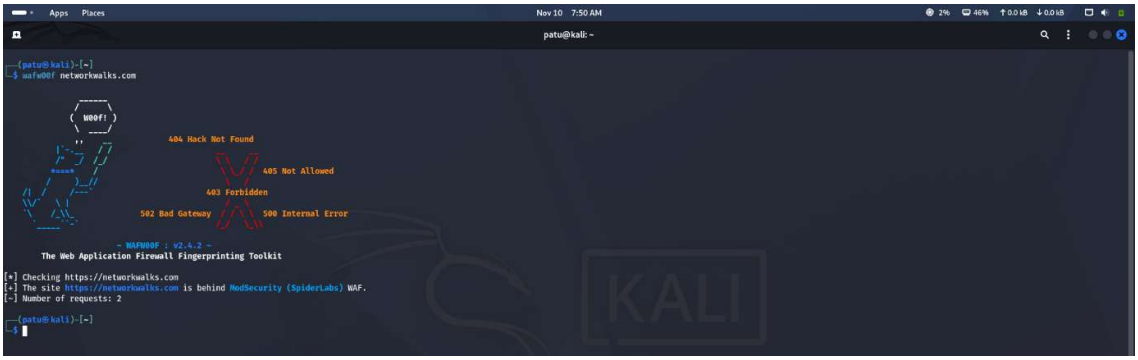
3. cURL HTTP response header output.



4. Nslookup domain resolution output.



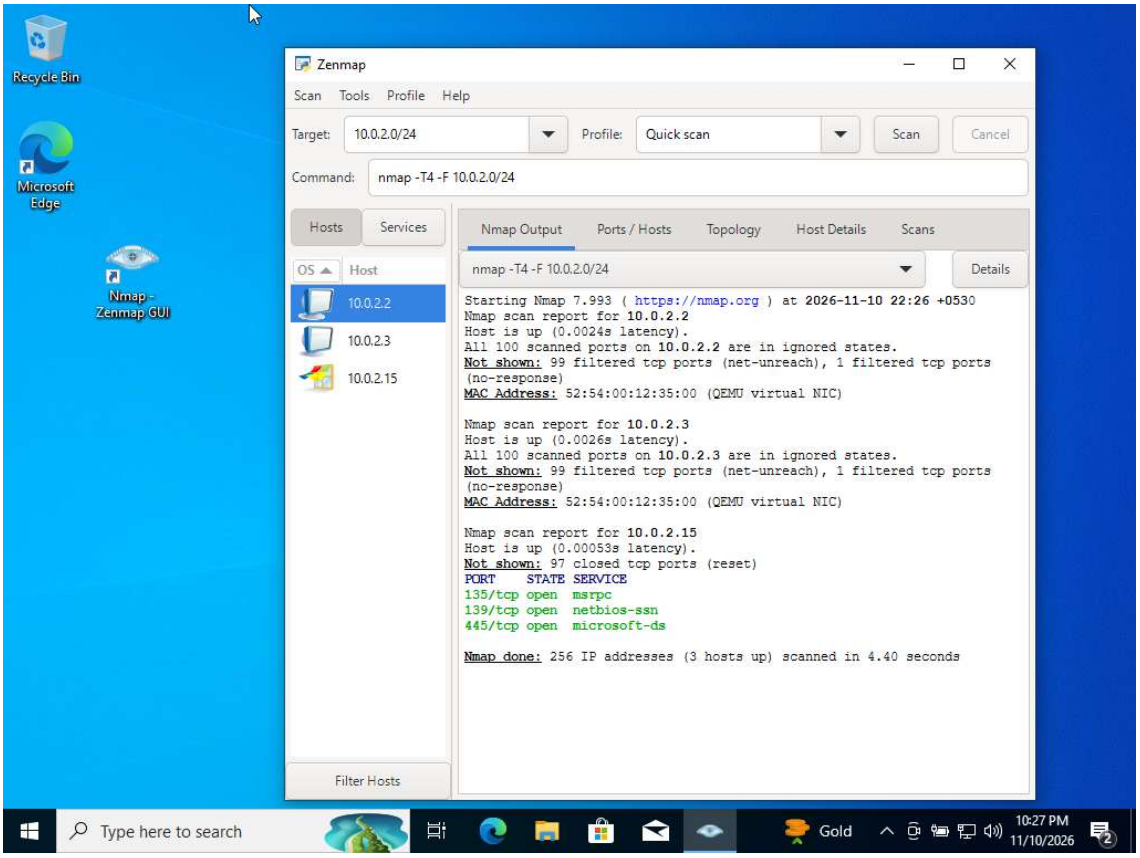
5. Wafw00f WAF detection result.

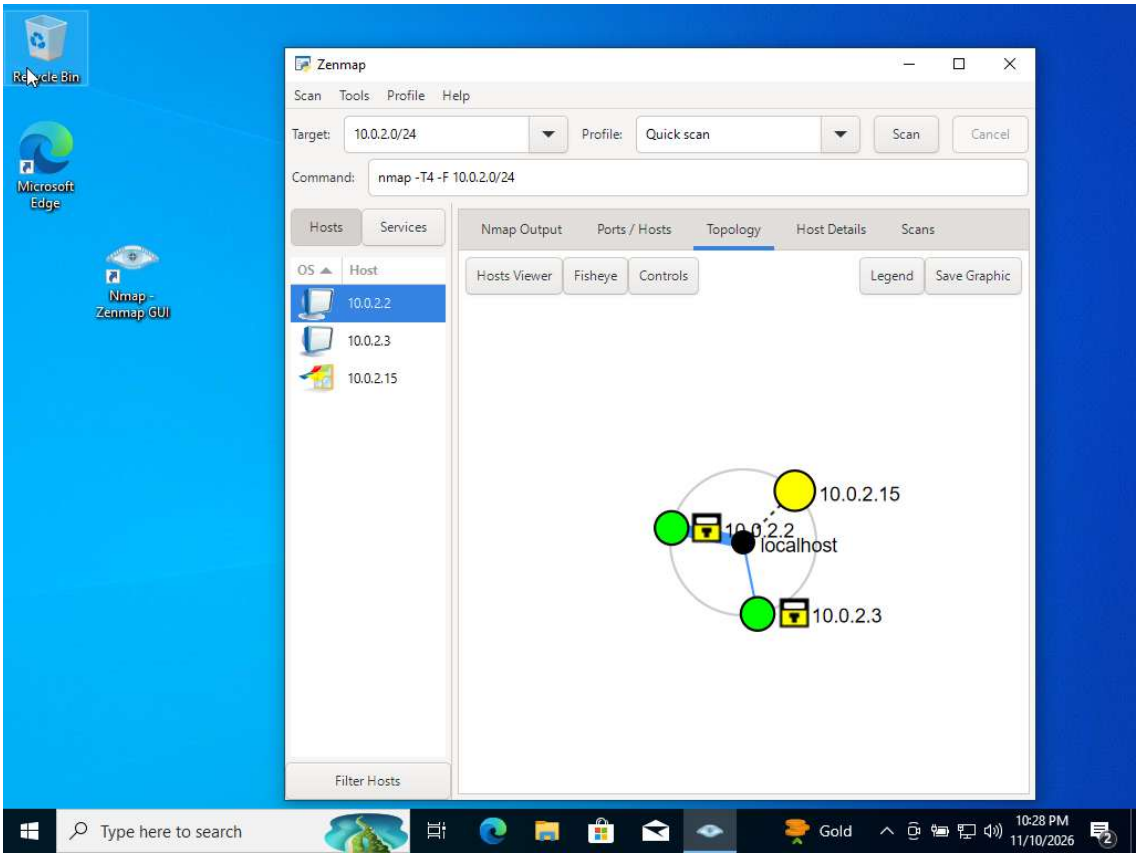


6. WhatWeb technology identification output.



7. Zenmap Scanning





END OF REPORT